

Miguel ALVES FERREIRA

📍 Room 10.26, Instituto Superior Técnico, Torre Norte, Lisboa
✉️ maferrei@andrew.cmu.edu

January 2026

RESEARCH INTERESTS

I am interested in the application of theoretical computer science to computer networking, namely to problems in internet routing, traffic engineering and network design. My PhD thesis investigates the important and timely problem of utilizing data-center networks fairly and efficiently having routing as the main design variable.

EDUCATION

PHD. ELECTRICAL AND COMPUTER ENGINEERING • *Carnegie Mellon University (CMU)* | *Instituto Superior Técnico (IST)* • 2021 - Present
Advised by [Justine Sherry](#) and [João Luís Sobrinho](#)

MSC. ELECTRICAL AND COMPUTER ENGINEERING • *Instituto Superior Técnico* • 2020
Major in Telecommunications and Computer Systems • Average Grade of 18 marks out of 20
Advised by [João Luís Sobrinho](#)
Thesis: [Routing on Multiple Optimality Criteria: Theory and Protocols](#)
★ Graded 20 marks out of 20. [IEEE Portugal Outstanding Master Thesis Award](#)

PUBLICATIONS

- [Miguel Ferreira](#), Nirav Atre, Justine Sherry, Michael Dinitz, and João Luís Sobrinho. [Minimum Congestion Routing of Unsplittable Flows in Data-Centers](#). In submission (2025)
- [Miguel Ferreira](#), Nirav Atre, Justine Sherry, and João Luís Sobrinho. [Impossibility Results for Data-Center Routing With Unsplittable Flows and Congestion Control](#). In Proceedings of the ACM Symposium on Principles of Distributed Computing 2024 (**PODC 2024**).
- João Luís Sobrinho, and [Miguel Ferreira](#). [From Non-Optimal Routing Protocols To Routing on Multiple Optimality Criteria](#). *IEEE/ACM Transaction on Networking* (**TON 2022**).
- João Luís Sobrinho, and [Miguel Ferreira](#). [Routing on Multiple Optimality Criteria](#). In Proceedings of the ACM Special Interest Group on Data Communication (**SIGCOMM 2020**).
★ [Best Paper Award](#)

TALKS

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| Minimum Congestion Routing on Data-Center Networks: | |
| • at IST Distributed, Parallel and Secure Systems Group | September 2025 |
| • at NSF Workshop on Reconfigurable Networks | June 2025 |
| Impossibility Results for Data-Center Routing: | |
| • at PODC 2024 | June 2024 |
| • at UT-Austin Systems and Networking Group | June 2024 |
| • at IST Distributed, Parallel and Secure Systems Group | May 2024 |
| • at CMU-Portugal Doctoral Symposium | Dec 2023 |
| Routing on Multiple Optimality Criteria: | |
| • at IST Distributed, Parallel and Secure Systems Group | Oct 2020 |
| • at SIGCOMM 2020 | August 2020 |

AWARDS

Student Travel Grant PODC 2024 ACM	2024
Dual Degree PhD Fellowship CMU Portugal	2021 - 2026
Outstanding Master Thesis Award IEEE Portugal	2020
Best Paper Award at SIGCOMM 2020 ACM	2020
Academic Excellence Certificate IST,	All semesters 2015 - 2020

TEACHING EXPERIENCE

Graduate Teaching Assistant Computer Networks and the Internet	IST, Spring 2026
Guest Lecturer Network Algorithms and Applications	IST, Fall 2025
Graduate Teaching Assistant Computer Networks and the Internet	IST, Spring 2025
Graduate Teaching Assistant Advanced Probability and Statistics	CMU, Spring 2023

MENTORSHIP EXPERIENCE

• Sabrina Hatch , CMU REUSE	CMU, Summer 2023
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RESEARCH PROJECTS

Towards Congestion-Free Data-Center Networks.	2024 - 2025
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The previous parts reveal that, if Clos networks wish to emulate a macro-switch, then further design variables beyond single-path routing and arbitrary rate allocation at each routing are required. In the third part of the thesis, which is under development, we are investigating two possible extra design variables. The first is variable virtual machine placement. Most applications are hosted in virtual machines, and cloud providers can choose in which servers to place each virtual machine. Accordingly, an idea is to jointly assign these virtual machines to servers and route the flows between them. The second is multi-path routing over a few paths. While arbitrary flow splitting is unrealistic, future transport protocols may tolerate a limited degree of splittability. Therefore, another idea is to jointly route flows and divide their demands over a small number of paths.

Minimum Congestion Routing in Data-Center Networks	2023 - 2024
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The second part of my thesis work maintains unsplittable flows but waves congestion control, and analyzes again the discrepancies between Clos networks and a macro-switch. Since data-centers are proprietary, cloud providers are not bound to congestion control, and, instead, they may support arbitrary rates at each routing. In this context, my key finding is that the scale-down factor by which Clos networks approximate a macro-switch is (in the worst-case) between $\frac{3}{2}$ and $\frac{9}{5}$. Notably, showing that this factor is at most $\frac{9}{5}$ entails designing a new routing algorithm that breaks through the factor of 2 implicit in existing heuristics. That is, while Clos networks fail to behave as a macro-switch under unsplittable flows, suitable approximation algorithms allow for a reasonable approximation.

Impossibility Results for Data-Center Routing	2022 - 2023
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Clos networks are the building block of data-centers. With arbitrarily splittable flows, or sources and destinations restricted to at most one flow, Clos networks have been shown to emulate a macro-switch connecting all sources to all destinations, such that flow rates are limited only by the link capacities at the ingress and egress points. Therefore, Clos networks are an attractive design for a range of use cases, and have been widely deployed. However, today's data-centers operate on different traffic premises: flows are unsplittable, and sources and destinations are not restricted to at most one flow, with congestion control imposing that link capacities are shared among flows according to max-min fair allocations.

The first part of my thesis work assumes unsplittable flows, and characterizes the effect

of congestion control on the discrepancies between Clos networks and a macro-switch. In my main result, I prove that there are sets of flows for which the max-min fair rates in the macro-switch cannot be replicated; in fact, I prove that, surprisingly, optimizing the routing for max-min fairness reduces the max-min fair rates in the macro-switch of some flows by a factor of proportional to the size of the network. In other words, routing for max-min fairness fails to closely replicate a macro-switch abstraction for fairness.

**Inter-AS Routing meets Multiple Optimality Criteria, and
On the Convergence Rates of Routing Protocols**

2020 - 2021

We extend the work described below in two directions. First, we apply the protocols design to overcome the limitation of BGP in achieving optimal goals for inter-domain routing, and how they open up the range of policy choices available. Second, we analyze the convergence rates of several types of vectoring protocols; namely, we determine their worst-case time and message complexity under synchronous and asynchronous regimes for different network events and attributes, with a special emphasis on vectoring protocols with diffusing updates, as substantiated by EIGRP.

Routing on Multiple Optimality Criteria

2019 - 2020

Standard routing protocols are unable to find optimal paths simultaneously for several criteria; in fact, they only find optimal paths, separately, for criteria that satisfy good mathematical properties. Our work introduces a solution to routing optimally and concurrently on multiple criteria. This solution comprises: (i) the development of a new theory, which transforms a collection of criteria that do not satisfy good mathematical properties into a dominance criterion that does; (ii) the design of new routing protocols, which find the sets of dominant paths and allow forwarding packets along each of these paths; (iii) an evaluation by simulation of the protocols designed, which attests their good behavior.